Week 95: Count the unseen

Adult guide built from the problem files · Unpiloted. Marks, mixing, resets and timing unrehearsed.

The answers, hints and proofs are in the print guide, week-95-facilitator.pdf, whose source is guide/facilitator.tex.

Problems and routes

Problem	2–3	4–5	route
Three live rounds with a hidden bag	P1	P1	first
Smallest and larger bags for two lettered catches	P2		first
Two catches with no shared objects	P3		first
Smallest bags for two records		P2	first
A fraction-matching guess		P3	first
Six printed rounds: 8 or 16?	P4		first
Overlaps 0 and 3 from bags of 8 and 16	P5		return
Five rounds, five guesses, one bag		P4	return
Catch sizes that force an overlap		P5	return
Every possible overlap for three bags	P6		return
Invent a record your bag could never give	P7		return
Split a handling budget of 12		P6	return
A rule for any budget		P7	return
Did lost marks cause the small overlap?		P8	return

First route. Grades 2–3 P1–P4; Grades 4–5 P1–P3. **Return visits.** Grades 2–3 P5–P7; Grades 4–5 P4–P8.

Key, one entry per problem

Three live rounds with a hidden bag

Grades 2–3 P1 · Grades 4–5 P1 · first route · wording differs by band

Idea. One unchanged bag gives different overlaps in different rounds; a guess is an inference, not a recovered total.

Smallest and larger bags for two lettered catches

Grades 2–3 P2 · first route

Idea. Two catches show exactly $M + n - R$ different objects, so that is the smallest bag, and any larger bag also fits.

Two catches with no shared objects

Grades 2–3 P3 · first route

Idea. Disjoint catches of 5 need at least 10 objects, and no record gives a largest bag.

Smallest bags for two records

Grades 4–5 P2 · first route

Idea. A record proves $N \geq M + n - R$ and no upper bound.

A fraction-matching guess

Grades 4–5 P3 · first route

Idea. Matching the marked fraction gives the estimate Mn/R, which a different possible bag shows is not forced.

Six printed rounds: 8 or 16?

Grades 2–3 P4 · first route

Idea. Every printed overlap fits both bags; the rounds lean one way without deciding.

Overlaps 0 and 3 from bags of 8 and 16

Grades 2–3 P5 · return route

Idea. Arranged catches prove what can happen; waiting for a rare random draw does not.

Five rounds, five guesses, one bag

Grades 4–5 P4 · return route

Idea. The rule's guesses vary from round to round while one bag of 18 fits every record; the zero round proves only $N \geq 12$.

Catch sizes that force an overlap

Grades 4–5 P5 · return route

Idea. In a bag of N, overlap 0 is impossible exactly when $M + n > N$; for every bag from 12 to 20 that needs catches of at most 12 totalling at least 21.

Every possible overlap for three bags

Grades 2–3 P6 · return route

Idea. In a bag of N the overlap takes every value from $\max(0, M + n - N)$ to $\min(M, n)$, so a big enough catch forces a mark.

Invent a record your bag could never give

Grades 2–3 P7 · return route

Idea. Knowing the overlap range of a bag lets a child build and check an impossible record.

Split a handling budget of 12

Grades 4–5 P6 · return route

Idea. With $M + n$ fixed, the average overlap Mn/N is greatest for equal catches.

A rule for any budget

Grades 4–5 P7 · return route

Idea. For every budget s the best split is $\lfloor s/2 \rfloor$ and $\lceil s/2 \rceil$.

Did lost marks cause the small overlap?

Grades 4–5 P8 · return route

Idea. Lost marks can only lower the detected overlap; letters give an independent check.